VFD Simulation and Analysis

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November 5, 2025

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Aim and Objectives

- To control the speed and torque of an AC induction motor by varying the reference frequency of PWM and voltage in a controlled manner.
- Design and simulate a Variable Frequency Drive (VFD) model using MATLAB Simulink.
- Understand the operation of each control and inverter block involved in AC motor speed control.

System Overview

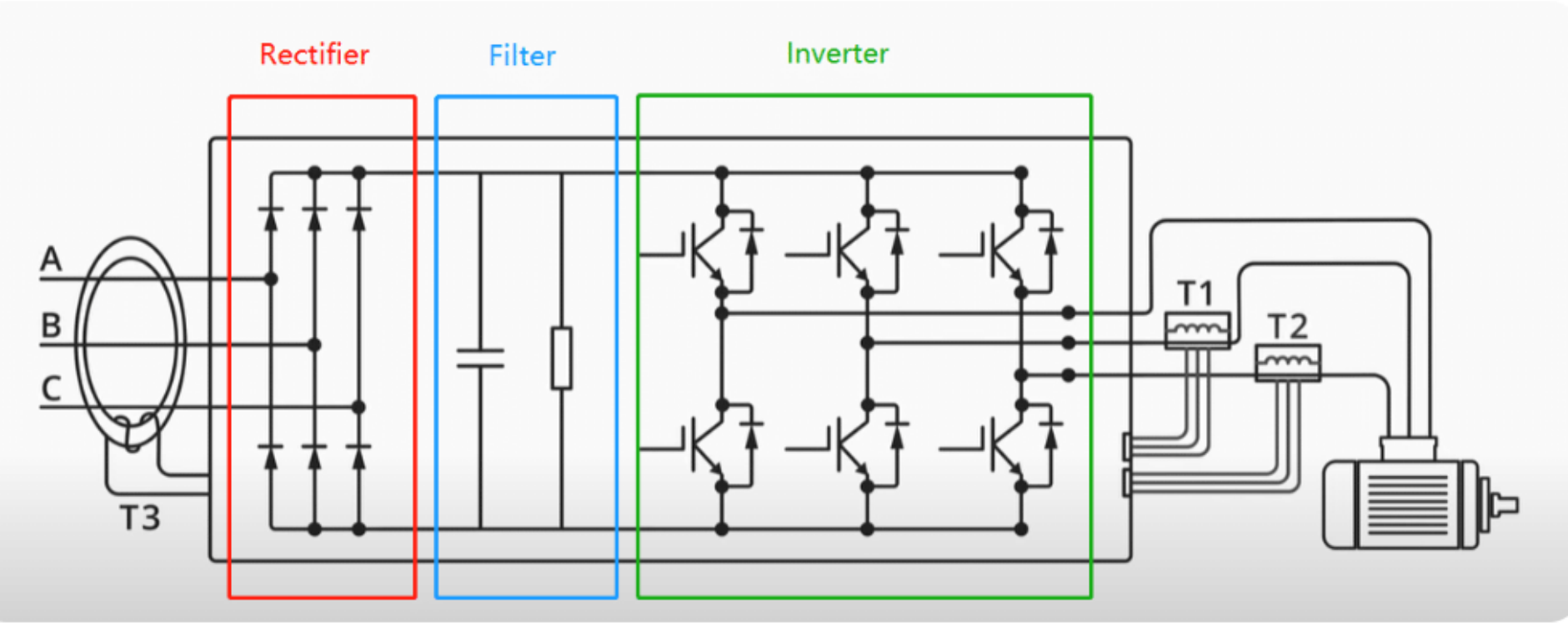


Figure: Power Flow Overview

Theory of Operation

- VFD supplies variable frequency and voltage to an induction motor.
- Inverter converts DC to AC using PWM.
- Motor speed equation: $N = \frac{120 \cdot f}{P}$
- Constant torque maintained by keeping V/f ratio constant.

Diode Bridge Rectifier

- Used instead of controlled rectifier for non-regenerative standard VFDs.
- Blocks reverse power flow during braking.
- Braking resistors needed to dissipate excess energy.

DC Link Section

- Provides stable DC input to inverter.
- Implemented using ideal DC source or filtered DC bus.
- A braking resistor is used in a VFD to dissipate excess energy during rapid deceleration, preventing overvoltage faults and protecting the drive.

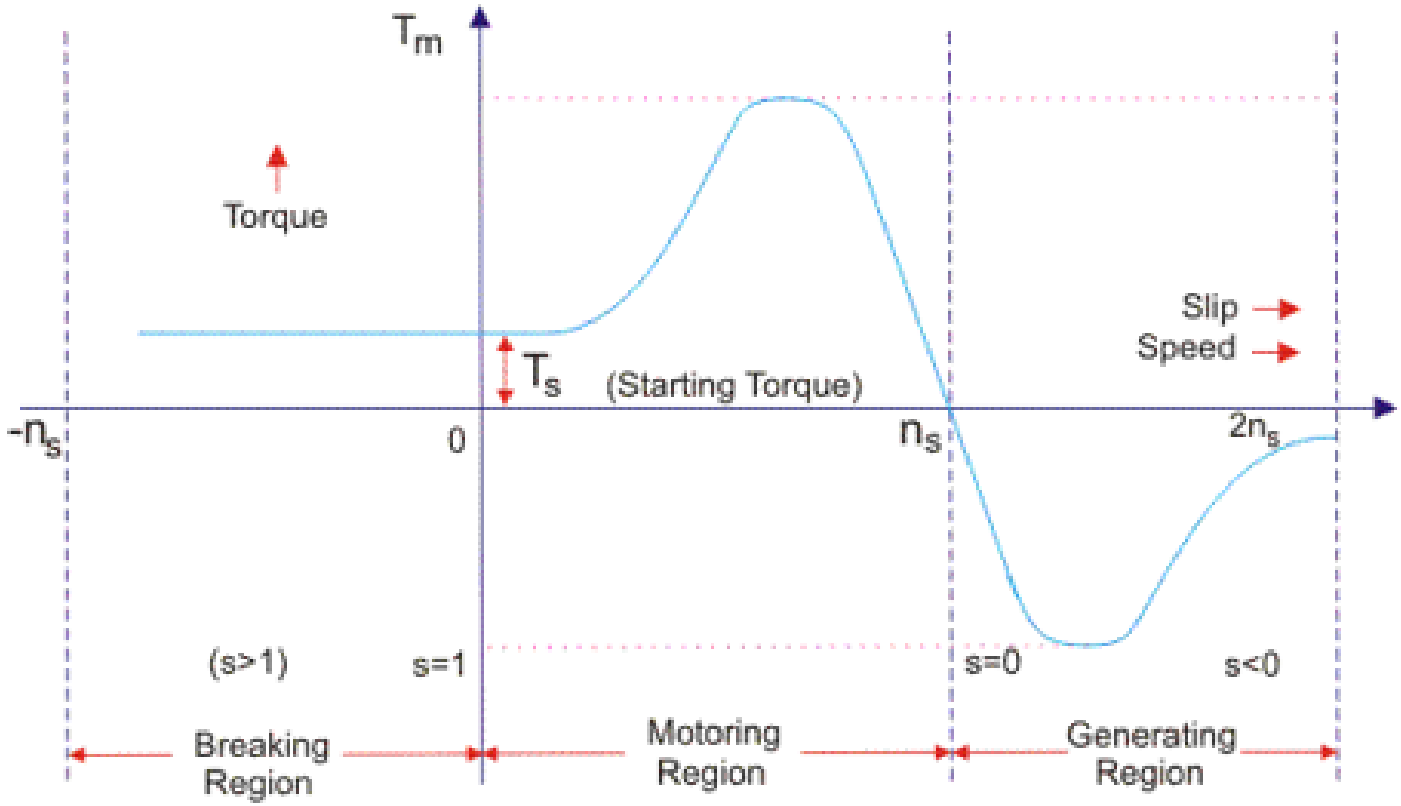
PWM Inverter

- Built using six IGBT switches in 3-phase bridge configuration.
- Controlled by SPWM signals for speed control.
- Reference signal amplitude controls frequency and thus motor speed.

Torque-Slip Characteristics

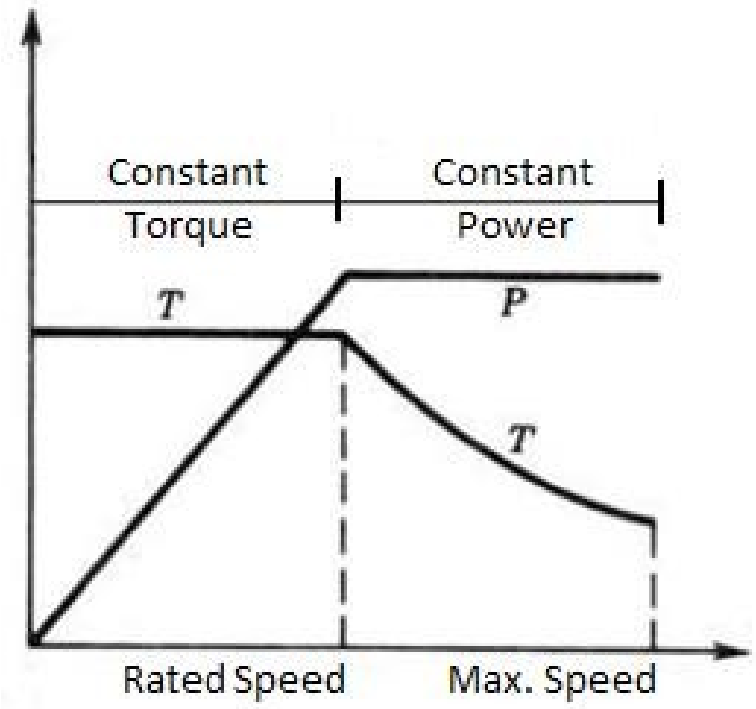
- Having three regions: low slip, medium slip, and high slip.

The curve can be described in three modes of operation-



Torque Slip Curve for Three Phase Induction Motor

VFD Characteristics



- Up to the rated speed: V/f = constant, power increases linearly, torque remains constant.
- At lower speed, reduced cooling is compensated by reduced heating due to lower power.

Design Parameters

Motor Ratings

- 3-Phase, 460 V
- 5 HP
- 1750 RPM

Torque Calculation:

$$P_{out} = T_{shaft} \times \omega$$

$$\therefore \quad T_{shaft} = \frac{746 \times 5}{2 \times \pi \times 1750}$$

Rated Torque:

$$T_{shaft} = 20.354 \text{ Nm}$$

VFD Ratings for Different Load Applications

460V Series Power [HP]	0.75 [1]	1.5 [2]	2.2 [3]	3.7 [5]	5.5 [7.5]	7.5 [10]	11 [15]	15 [20]	18.5 [25]	22 [30]	30 [40]	37 [50]	45 [60]	55 [75]	75 [100]
Model Code	5	7	9	11	13	15	17	19	21	23	25	27	29	31	33
Rated Current of the Fixed Torque	3	4.2	6	8.5	13	18	24	32	38	45	60	73	91	110	150
Rated Current of the Variable Torque	3.75	5.25	7.5	10.625	16.25	22.5	30	40	47.5	56.25	75	91.25	113.75	137.5	187.5
The Greatest Carrier Wave Frequency	15KHz										10KHz				6KHz

Figure: VFD ratings for different load applications

Braking Resistor Calculations I

A braking resistor is used in a VFD to dissipate the excess energy generated when a motor is rapidly decelerated. Without it, this energy is fed back into the VFD's DC bus, causing the voltage to rise to a dangerous level, which can damage the drive and trigger an overvoltage fault. The resistor safely converts this regenerated electrical energy into heat.

$$P_{dc} = V_{dc} \times I_{dc}$$

$$R_{braking} = \frac{V_{dc}^2}{P_{dc}}$$

For selecting VFD rating for a particular motor:

$$VFD_{\text{power rating}} = 1.5 \times \text{Motor rating}$$

As we have a 3-phase induction motor (60 Hz, 460 V, 5 HP):

Braking Resistor Calculations II

$$VFD_{\text{power rating}} = 1.5 \times 5 = 7.5 \text{ HP}$$

$$VFD_{\text{power rating}} = 7.5 \times 746 = 5595 \text{ Watt}$$

$$V_{dc} = 1.414 \times 1.25 \times 460 = 813.05 \text{ Volts}$$

$$R_{braking} = \frac{V_{dc}^2}{P_{dc}} = \frac{813.05^2}{5595} = 118.1501 \ \Omega$$

Basic Coupling Capacitor Relation I

Relation:

$$I_{dc} = \frac{P_{dc}}{V_{dc}}$$

$$\therefore I_{dc} = \frac{5595}{650} \quad \cdots(1)$$

$$I_{dc} = 8.60\ A \quad \cdots(2)$$

Whereas Δt = ripple time:

$$\Delta t = \frac{1}{f_{ripple}}$$

$$f_{ripple} = 6 \times f_{ac}$$

$$\therefore f_{ripple} = 6 \times 60 = 360\ Hz$$

Basic Coupling Capacitor Relation II

$$\therefore \Delta t = \frac{1}{360} = 2.77\ ms$$

From equation (1) & (2):

$$C = \frac{I_{dc} * safetymargin}{6 \times f_{ac} \times \Delta V}$$

If we consider 5% ripple of V_{dc}:

$$\therefore C = \frac{8.60 * 1.5}{6 \times 60 \times 32.5}$$

$$C = 881.50\ \mu F$$

System Single Line Diagram

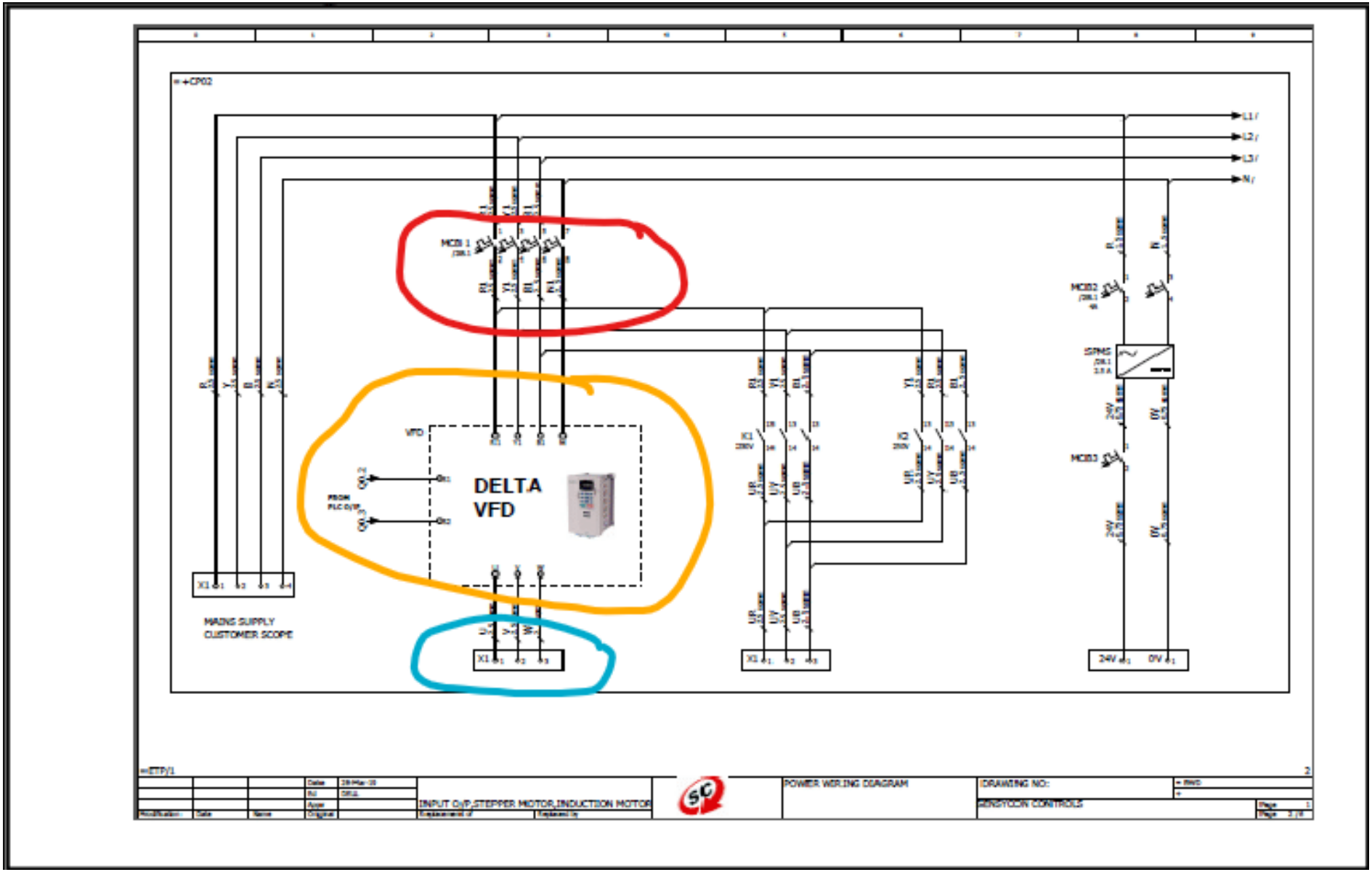


Figure: System single line diagram

References

- VFD data: Delta Electronics VFD V-series user manual.
- Basic concepts of the induction motor and power electronics have been taken from class notes.

Conclusion

- VFD enables precise control of speed and torque.
- SPWM technique ensures smooth motor operation.
- Energy savings and improvement in reliability.
- Helps in power factor improvement of Three phase Induction motor.